TwinRead

Two pathology foundation models as two independent readers: fused for accuracy, flagged for review, and shrunk to run offline on an Arduino UNO Q.

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Two models, trained differently, miss different tiles

- **CONCH** (Lu et al., *Nat Med* 2024): contrastive image–caption pretraining on 1.17 million histology image–caption pairs.
- **MUSK** (Xiang et al., *Nature* 2025): masked modelling on 50 million pathology images and 1 billion text tokens, unpaired.
- Different data and different objectives give different blind spots. On the 7,180 CRC-VAL-HE-7K test tiles, CONCH misses 329 and MUSK 286, but only 110 are missed by both.
- Each reader gets most of the other's misses right: 219 of CONCH's 329 errors (67%) and 176 of MUSK's 286 (62%). That is the gap a second reader can cover.

Tiles that at least one reader gets wrong (505 of 7,180)



Linear probes on frozen features, CRC-VAL-HE-7K, 9 tissue classes.
Source: results/cm_local/CM/crc/reader_error_overlap.json

Fusing the two readers leaves 239 errors, fewer than either reader alone.

Aim: a second read on every tile, then make it run anywhere

1

Two readers, one call

Treat CONCH and MUSK as two independent readers of every H&E tile and fuse them, so each covers the other's gaps.

2

Know when to ask a pathologist

Within a fixed review budget, send the tiles the fused reader is least sure about for a human second look. Count the blind spots both readers share.

3

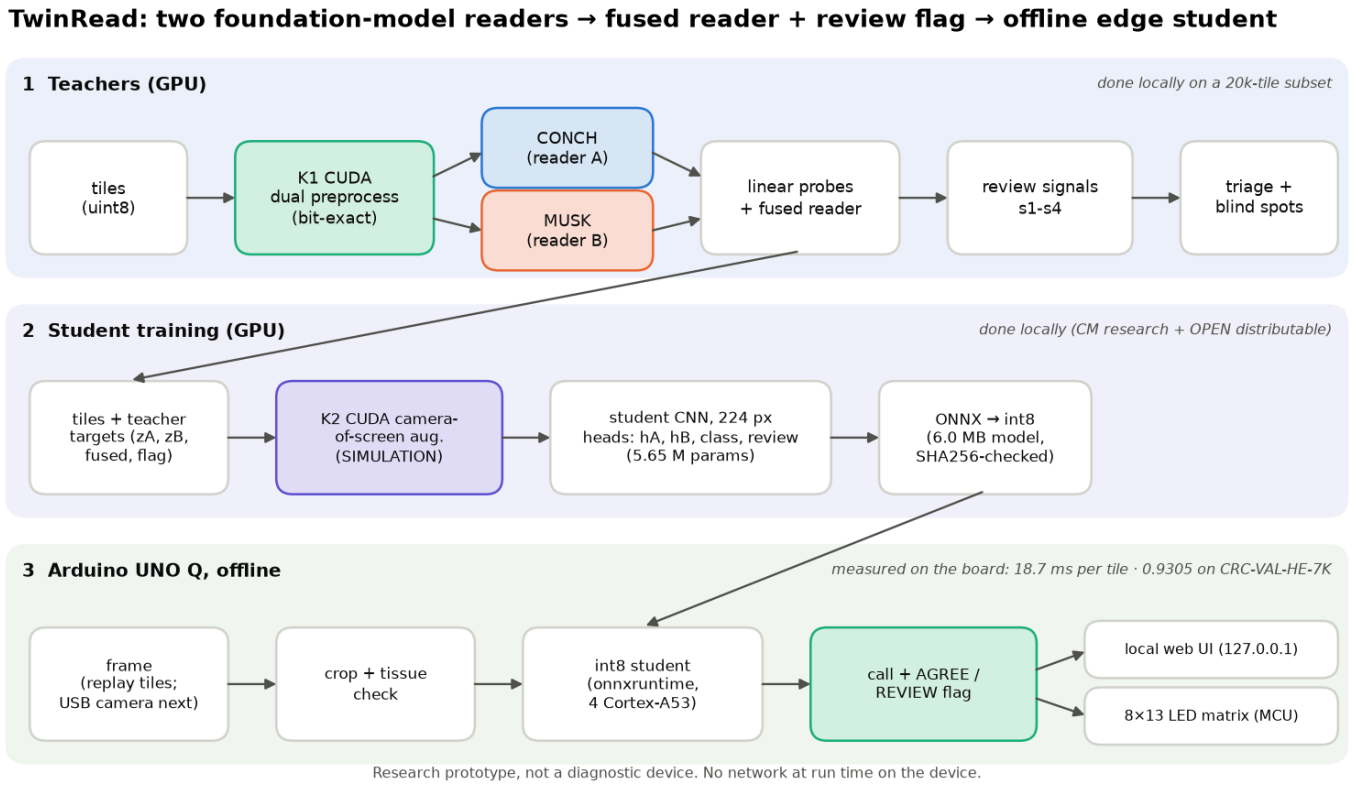
Run it with no internet, no GPU, no mains power

Distil both readers into one 6 MB int8 model and run it on a USB-powered Arduino UNO Q, for settings with no scanner, no server and no reliable connectivity.

Every design choice (fusion rule, review threshold) is made on validation data and pre-registered before any test label is read.

Method

- **Readers.** A linear probe on frozen CONCH features and one on frozen MUSK features. A custom CUDA kernel (K1) prepares both readers' inputs from one tile batch, bit-exact to the official transforms.
- **Fusion and review.** Three fusion candidates; the winner is chosen on validation. Review signals: disagreement, Jensen–Shannon divergence, fused uncertainty, and a learned combination.
- **Student.** A 5.65 M-parameter network learns both readers' embeddings, the fused call and the review label, trained with a GPU augmentation kernel (K2), then exported to int8 ONNX.
- **Data.** NCT-CRC-HE-100K (20k-tile training subset), CRC-VAL-HE-7K (test), Camelyon17-WILDS (unseen hospital as test).



Fusion covers the gaps: +0.74 points on CRC

0.9413 CONCH alone

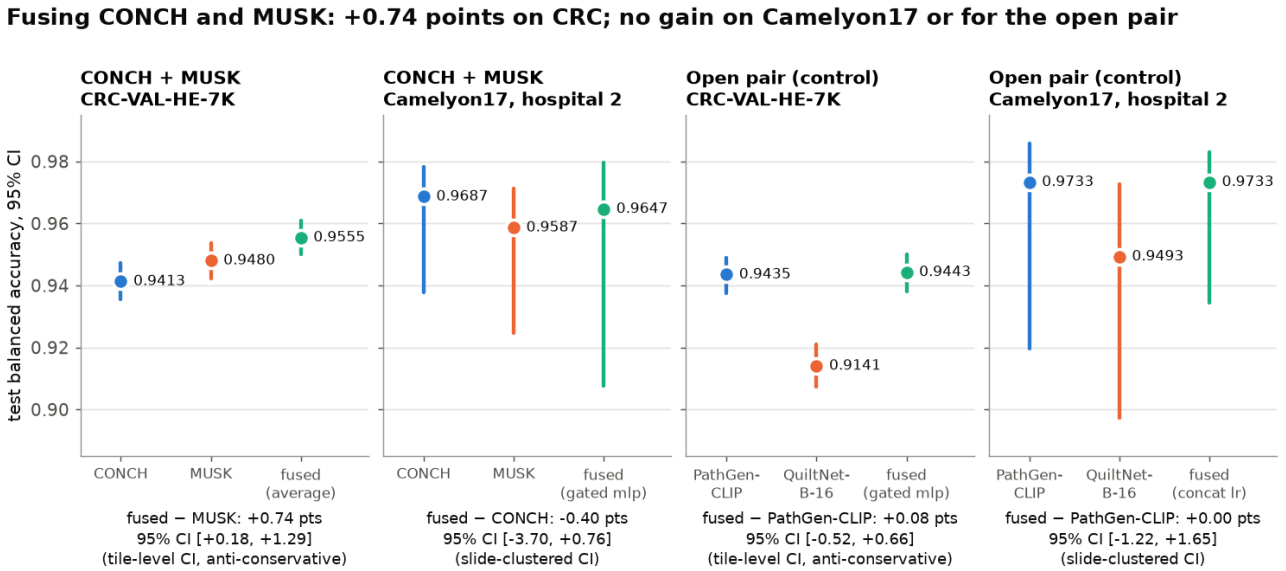
0.9480 MUSK alone

0.9555 CONCH + MUSK fused

Balanced accuracy, CRC-VAL-HE-7K, 7,180 tiles, 9 classes.

+0.74 points over the best single reader (95% CI +0.18 to +1.29, tile-level). Most of the gain is tumour recall, +3.6 points.

Where it does not help: on Camelyon17 (tumour vs normal, unseen hospital) CONCH alone is already at 0.9687 and fusion gives no gain (0.9647). An open-model control pair gains nothing on either dataset. The gain is real but small and dataset-dependent.



Linear probes on frozen features. The fusion rule was chosen on validation and pre-registered before test. CRC: train on a 20k-tile NCT-CRC-HE-100K subset, test on CRC-VAL-HE-7K (7,180 tiles, no slide ids). Camelyon17-WILDS: test on unseen hospital 2 (1,500 tiles). Local-subset numbers (20k of 100k CRC training tiles). On CRC, large opposing MUS/STR recall swings cancel; the net gain is mostly tumour recall (+3.6 pts). Validation picked the fusion rule by 1-2 tiles. Research prototype, not a diagnostic device.

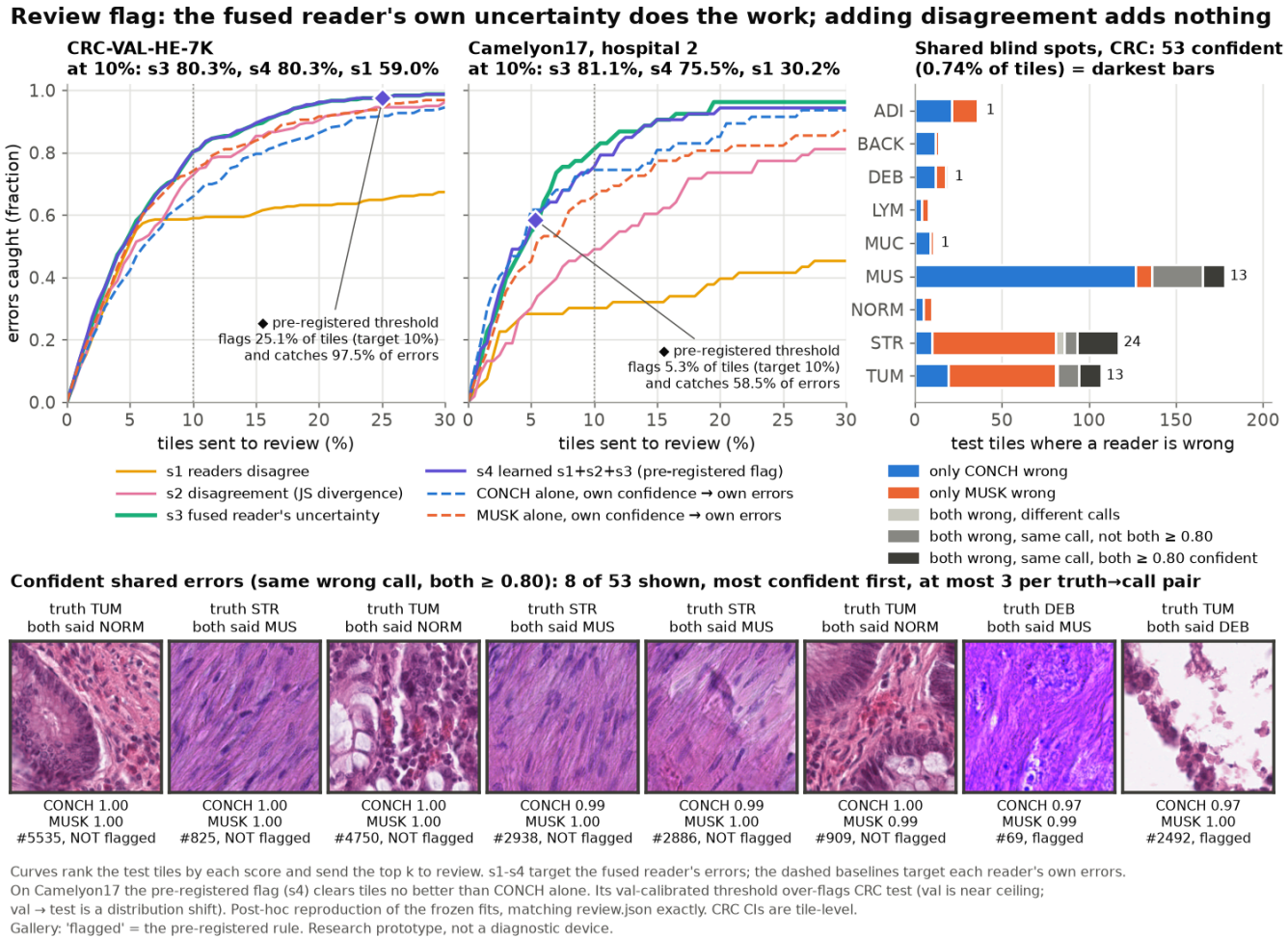
The fused reader knows which tiles to send for review

80.3%

of the fused reader's errors are caught by reviewing the 10% of tiles it is least sure about

The same budget catches 74.1% with MUSK checking itself and 66.0% with CONCH. Disagreement between the two readers alone catches 59.0%: the value comes from a better-calibrated fused reader, not from disagreement itself.

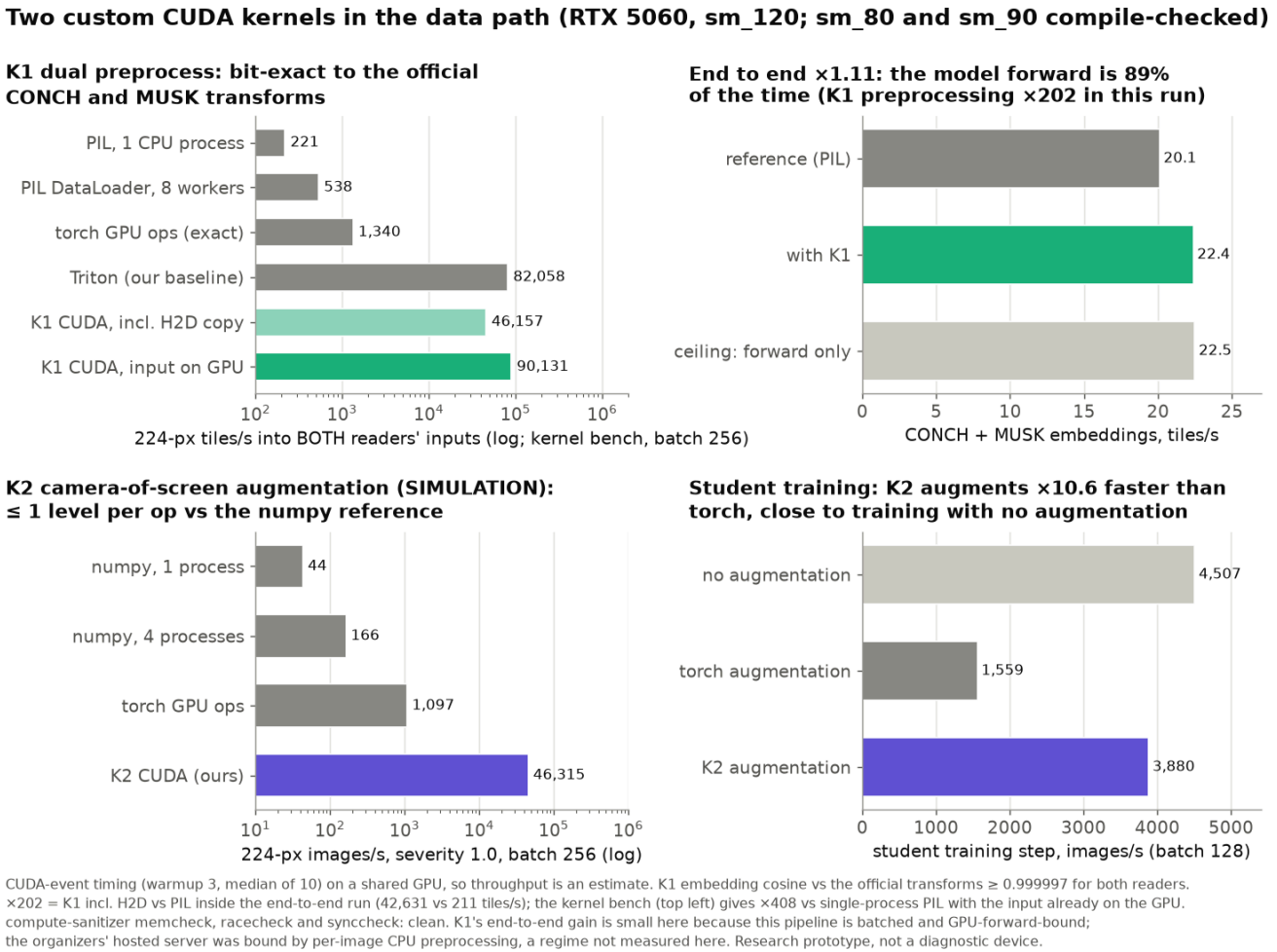
Shared blind spots: 53 tiles (0.74%) are called confidently wrong by both readers, mostly muscle vs stroma. No disagreement signal can catch those.



Curves rank the test tiles by each score and send the top k to review. s1-s4 target the fused reader's errors; the dashed baselines target each reader's own errors.
On Camelyon17 the pre-registered flag (s4) clears tiles no better than CONCH alone. Its val-calibrated threshold over-flags CRC test (val is near ceiling; val → test is a distribution shift). Post-hoc reproduction of the frozen fits, matching review.json exactly. CRC CIs are tile-level.
Gallery: 'flagged' = the pre-registered rule. Research prototype, not a diagnostic device.

Two CUDA kernels make the pair cheap to run and to train

- **K1, dual-reader preprocessing.** One launch turns a uint8 tile batch into both readers' inputs (448 px and 384 px), bit-exact to the official transforms. 90,131 tiles/s on an RTX 5060; preprocessing ×202 faster than PIL inside the embedding run, ×1.11 end to end because the forward pass dominates.
- **K2, training augmentation.** A simulated camera-of-screen degradation (warp, stain jitter, moiré, glare, defocus, noise, JPEG) at 46,315 images/s; student training ×10.6 faster than torch augmentation.
- Verified against numpy references and compute-sanitizer; built for sm_80, sm_90 and sm_120.



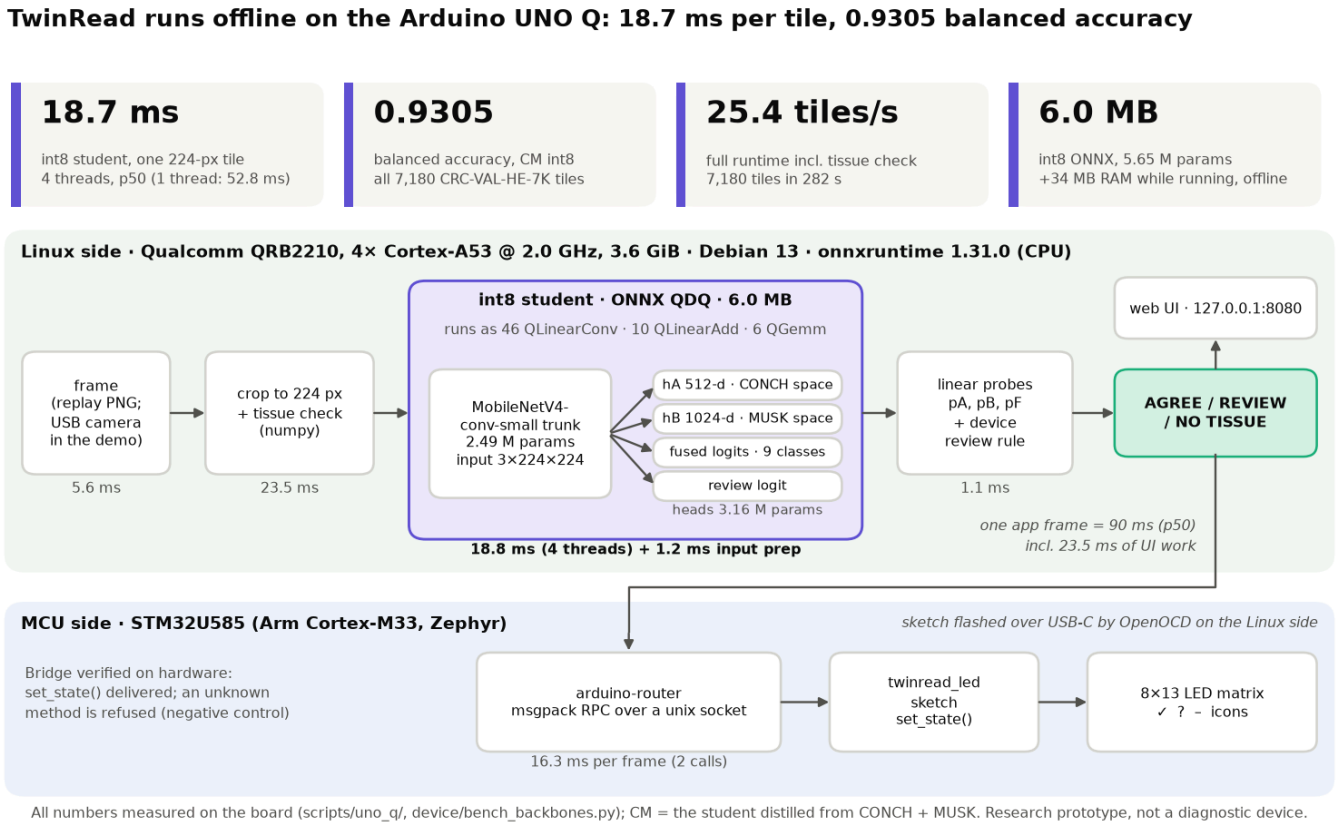
CUDA-event timing (warmup 3, median of 10) on a shared GPU, so throughput is an estimate. K1 embedding cosine vs the official transforms ≥ 0.999997 for both readers. ×202 = K1 incl. H2D vs PIL inside the end-to-end run (42,631 vs 211 tiles/s); the kernel bench (top left) gives ×408 vs single-process PIL with the input already on the GPU. compute-sanitizer memcheck, racecheck and synccheck: clean. K1's end-to-end gain is small here because this pipeline is batched and GPU-forward-bound; the organizers' hosted server was bound by per-image CPU preprocessing, a regime not measured here. Research prototype, not a diagnostic device.

It runs on the board

18.7 ms
per tile, int8 student, 4 threads

0.9305
balanced accuracy on all 7,180 CRC test tiles

6.0 MB
model, 34 MB RAM while running, no network

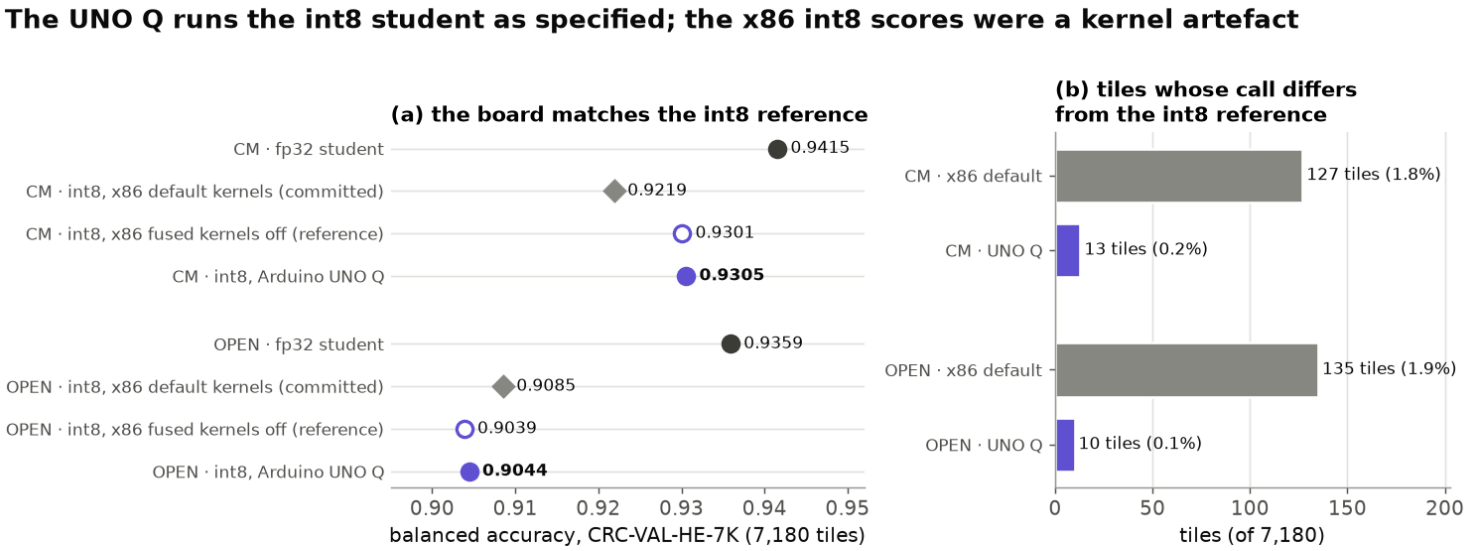


Arduino UNO Q: four Cortex-A53 cores, 3.6 GiB RAM, USB-powered, Debian 13, onnxruntime on CPU.

The full app loop (frame, tissue check, student, flag, LED) processes 25.4 tiles/s. The LED-matrix sketch is flashed and the Bridge RPC to the MCU is verified.

Every number on this slide was measured on the board.

The board runs the model as specified



Same model and input bytes on both machines (SHA-256). x86 = AMD Ryzen 7 5800X3D 8-Core Processor (AVX2, no VNNI), onnxruntime 1.31.0 with its default fused int8 kernels.
Reference = the same x86 with graph optimisations off, so QDQ runs as dequantise → fp32 op → quantise. Source: scripts/uno_q/.
Review flag vs the teachers' flag (kappa): CM 0.32 on the board vs 0.34 on x86, OPEN 0.29 vs 0.31; the bundle thresholds were calibrated on x86-kernel outputs.
Research prototype, not a diagnostic device.

On x86, onnxruntime's default fused int8 kernels scored the CM student at 0.9219. The board gives 0.9305, and so does x86 with those kernels switched off: 13 of 7,180 calls differ between the board and that reference.

The board's review flag matches the teachers' flag at kappa 0.32. The device's two "readers" are two heads of one network, so they are less independent than CONCH and MUSK.

Challenges

Overcome

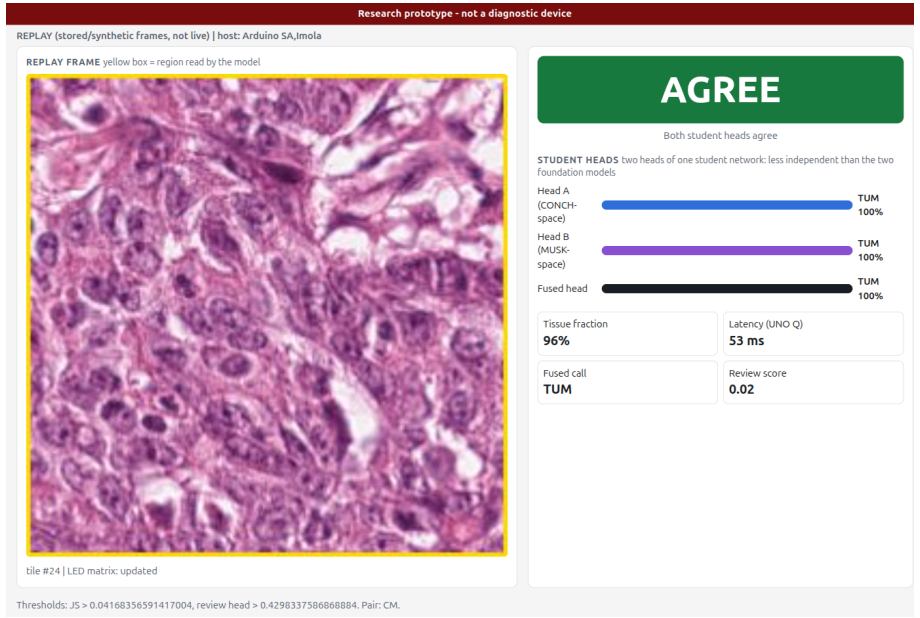
- CONCH and MUSK are CC-BY-NC-ND. Two configurations: the CONCH + MUSK student stays local, and an open pair (PathGen-CLIP + QuiltNet) is the distributable control.
- int8 calibration collapsed with MinMax and Entropy; percentile calibration fixed it.
- The x86 int8 scores were an onnxruntime kernel artefact, found only by running on the board.
- The pre-registered threshold over-flagged the test set (25.1% for a 10% target) because validation was near ceiling; reported as measured.

Remaining

- Real camera captures. The board has run on stored tiles; the screen-capture rig is ready.
- Full-data run on NCSA Delta: 100k training tiles instead of 20k, to see whether the fusion gain holds.
- Slide identifiers for CRC, so confidence intervals can be clustered by slide.
- Per-site threshold calibration, then clinical validation.

Why it matters

- **For pathologists:** a second read on every tile and a short list of tiles worth a second look, with both readers' calls shown, instead of a single black-box score.
- **For patients:** a cheap offline second look where there is no second pathologist, no scanner, no GPU, no internet and no reliable mains power. The whole system fits in 6 MB on a USB-powered board.
- **Limits:** the fusion gain is small and dataset-dependent; the device's two readers share one backbone; nothing here is clinically validated. Research prototype, not a diagnostic device.



The app on the UNO Q: a stored CRC tile, both student heads, the fused call and AGREE / REVIEW.

References

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Xiang J, Wang X, Zhang X, et al. A vision-language foundation model for precision oncology (MUSK). *Nature* 2025. doi:10.1038/s41586-024-08378-w

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Sun Y, Wu H, Zhu C, et al. PathMMU: a massive multimodal expert-level benchmark for understanding and reasoning in pathology. ECCV 2024. arXiv:2401.16355

Li et al. Multi-modal foundation models for computational pathology: a survey. 2025. PMC13081565

Muneer A, Waqas M, Saad MB, et al. From classical machine learning to emerging foundation models: review on multimodal data integration for cancer research. *Artificial Intelligence Review* 2026. doi:10.1007/s10462-026-11522-9

Kather JN et al. NCT-CRC-HE-100K and CRC-VAL-HE-7K (CC-BY-4.0). Bandi P et al. Camelyon17; Koh PW et al. WILDS (Camelyon17-WILDS, CC0).

Open models: PathGen-CLIP (jamessyx/pathgenclip-vit-large-patch14-hf, CC-BY-4.0), QuiltNet-B-16 (wisdomik/QuiltNet-B-16, MIT), timm mobilenetv4_conv_small.

Code, results JSON and figures: github.com/iujab/twinread-conch-musk-edge. Every number in this deck is read from results/headline.json or the JSON it names.